SCHOOL EXAMINATION · GRADE 11

AS Level Physics

Kinematics, Vectors and Electrostatic Force

DURATION	TOTAL	SECTIONS	MATERIALS
1 hour 15 minutes	50 marks	A · B · C	Calculator, ruler, pencil, protractor

CANDIDATE NAME	CLASS	DATE

INSTRUCTIONS

- Answer **all** questions in all three sections.
- **Section A:** for each question choose **one** answer, A, B, C or D, and circle it clearly.
- **Sections B and C:** write your answers in the spaces provided. Show all your working — marks are given for correct method.
- Give numerical answers to an appropriate number of significant figures, with units.
- You may use a calculator. Take $k = 8.99 \times 10^9$ N m^2 C^{-2}.
- The number of marks is given in brackets [ ] at the end of each question or part question.
- Diagrams are **not** drawn to scale unless stated.

Section	Paper style	Q	Max	Mark
A	Multiple choice (P1)	1 – 15	15	
B	Structured (P2)	16	6	
		17	6	
		18	6	
		19	7	
C	Practical skills (P3)	20	10	
Total			**50**	

Weighting: Section A 30% · Section B 50% · Section C 20%

DATA AND FORMULAE

Coulomb constant	$k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^9$ N m^2 C^{-2}
1 μC (microcoulomb)	1×10^{-6} C
Average speed / velocity	$v = s / t$
Acceleration	$a = (v - u) / t$
Uniformly accelerated motion	$s = ut + \tfrac{1}{2}at^2$
Components of a vector F at angle θ to the horizontal	$F_x = F\cos\theta$, $F_y = F\sin\theta$
Coulomb's law	$F = Q_1Q_2 / (4\pi\varepsilon_0 r^2)$

Section A · Multiple choice

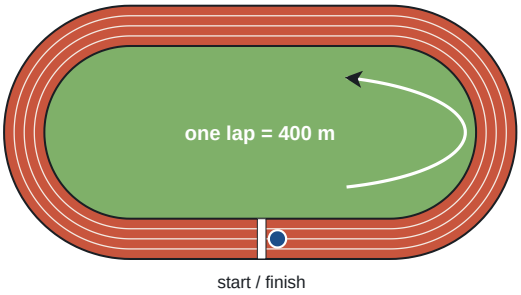
15 questions · 1 mark each · 15 marks

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and circle it.

1 Which quantity is a **vector**?

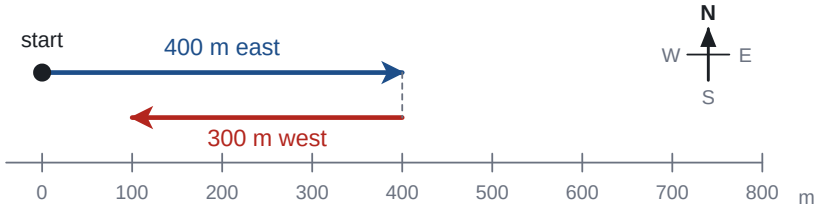
A distance **B** speed **C** time **D** displacement

2 An athlete runs exactly one lap of the 400 m track shown, starting and finishing at the same point. The lap takes 80 s.



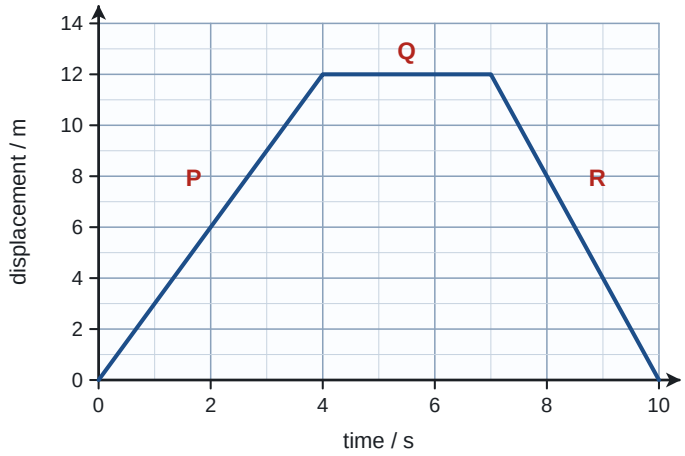
	average speed / m s^{-1}	average velocity / m s^{-1}
A	5.0	0
B	0	5.0
C	5.0	5.0
D	0	0

3 A student walks 400 m due east and then 300 m due west along the same straight path.



	total distance travelled	final displacement from start
A	100 m	700 m east
B	700 m	100 m east
C	700 m	700 m east
D	100 m	100 m east

4 The displacement–time graph shows the motion of a toy car along a straight track. It has three sections, P, Q and R.



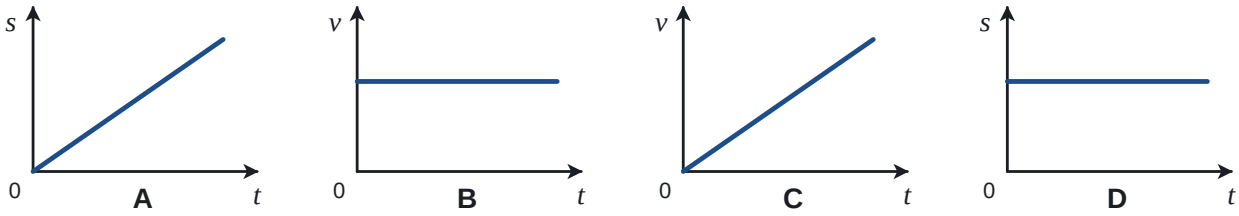
What is the velocity of the car during section R?

A −4.0 m s^{-1} **B** −3.0 m s^{-1} **C** 0 **D** +4.0 m s^{-1}

5 A car accelerates uniformly from 5.0 m s^{-1} to 25 m s^{-1} in 8.0 s. What is its acceleration?

A 0.40 m s^{-2} **B** 2.5 m s^{-2} **C** 3.1 m s^{-2} **D** 3.8 m s^{-2}

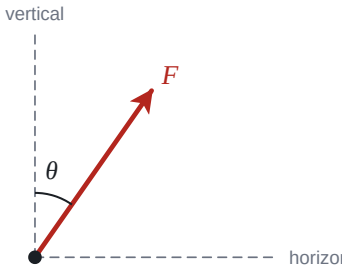
6 Which graph shows an object moving with a **constant, non-zero acceleration**? (s = displacement, v = velocity, t = time)



7 A car travels 30 km in 0.50 h and then a further 30 km in 1.0 h. What is its average speed for the whole journey?

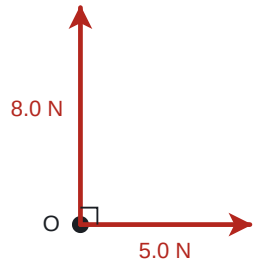
A 30 km h^{-1} **B** 36 km h^{-1} **C** 40 km h^{-1} **D** 45 km h^{-1}

8 A force F acts at an angle θ to the **vertical**, as shown.



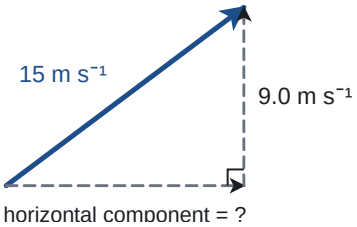
	horizontal component	vertical component
A	$F \sin \theta$	$F \cos \theta$
B	$F \cos \theta$	$F \sin \theta$
C	$F \tan \theta$	F
D	$F \cos \theta$	$F \cos \theta$

9 Two forces of 5.0 N and 8.0 N act at right angles on a point O. What is the magnitude of the resultant force?



A 3.0 N **B** 6.5 N **C** 9.4 N **D** 13 N

10 A ball moves with a velocity of 15 m s^{-1}. The vertical component of this velocity is 9.0 m s^{-1}. What is the horizontal component?



A 6.0 m s^{-1} **B** 12 m s^{-1} **C** 17.5 m s^{-1} **D** 24 m s^{-1}

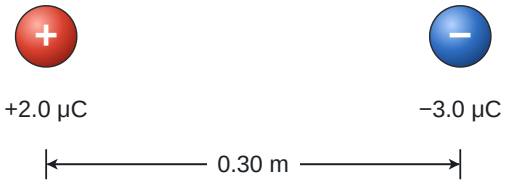
11 Two forces of 5.0 N and 12 N act at the same point. The angle between them can be changed. Which value **cannot** be the magnitude of their resultant?

A 7.0 N **B** 13 N **C** 17 N **D** 19 N

12 Two point charges exert an electrostatic force F on each other. The distance between them is doubled and the size of **one** of the charges is also doubled. What is the new force?

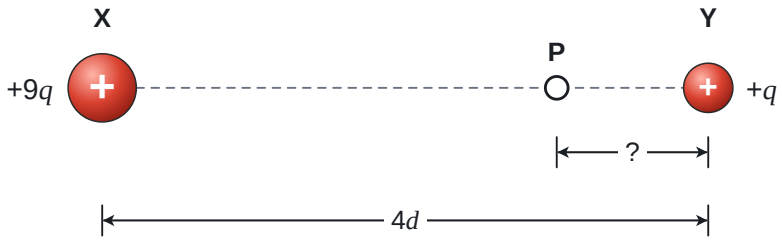
A $F/4$ **B** $F/2$ **C** F **D** $2F$

13 Two point charges are placed 0.30 m apart in air.



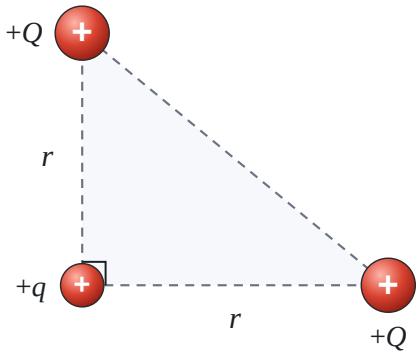
	magnitude of force on each charge	type of force
A	0.60 N	repulsive
B	0.18 N	attractive
C	0.18 N	repulsive
D	0.60 N	attractive

14 Two positive charges +9q and +q are fixed at X and Y, a distance 4d apart. A small positive charge placed at point P on the line XY experiences **zero** resultant force. How far is P from Y?



A 1.0d **B** 2.0d **C** 3.0d **D** 3.6d

15 Three charges are placed at the corners of a right-angled triangle. The two charges +Q are each a distance r from the charge +q. Each charge +Q on its own exerts a force of magnitude F on +q. What is the magnitude of the resultant force on +q?



A 0 **B** F **C** $\sqrt{2}\,F$ **D** $2F$

Section B · Structured questions

Answer all questions · 25 marks

Write your answers in the spaces provided. Show your working clearly in all calculations.

16

A cyclist rides from P to Q and then from Q to R along the two straight roads shown on the map in Fig. 16.1.

The road PQ points due east and the road QR points due north.

The distance and the time taken for each stage are shown on the map.

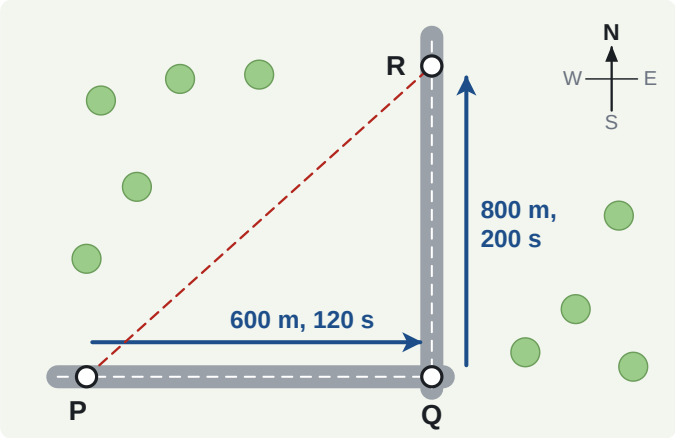


Fig. 16.1 (not to scale)

(a) State **one** difference between *distance* and *displacement*. **[1]**

.....

.....

(b) Calculate the total distance travelled by the cyclist from P to R.

distance = ... m **[1]**

(c) Calculate the magnitude and the direction of the cyclist's displacement from P to R. Give the direction as an angle measured from due east.

magnitude = ... m

direction = ... ° north of east **[2]**

(d) Calculate, for the whole journey from P to R,

(i) the average speed,

average speed = ... $m\ s^{-1}$ **[1]**

(ii) the magnitude of the average velocity.

average velocity = ... $m\ s^{-1}$ **[1]**

[Total: 6]

17 A train moves along a straight, level track between two stations. Fig. 17.1 shows how its velocity changes with time.

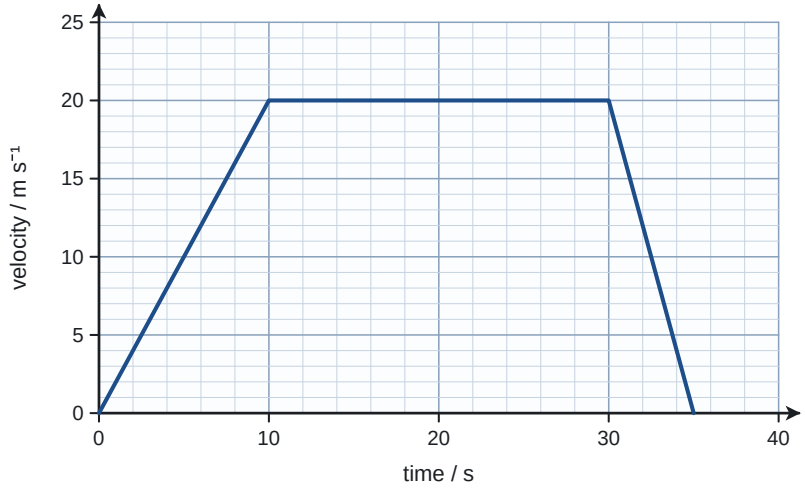


Fig. 17.1

(a) Define *acceleration*. **[1]**

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(b) Use Fig. 17.1 to calculate the acceleration of the train

(i) from time t = 0 to t = 10 s,

acceleration = ... m s^{-2} **[1]**

(ii) from time t = 30 s to t = 35 s.

acceleration = ... m s^{-2} **[1]**

(c) Determine the total displacement of the train in the 35 s journey.

displacement = ... m **[2]**

(d) Hence calculate the average velocity of the train for the journey.

average velocity = ... m s^{-1} **[1]**

[Total: 6]

18 **(a)** State the difference between a *scalar* quantity and a *vector* quantity. **[1]**

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(b) A child pulls a loaded sledge across level snow using a rope. The tension in the rope is 50 N and the rope makes an angle of 35° with the horizontal, as shown in Fig. 18.1.

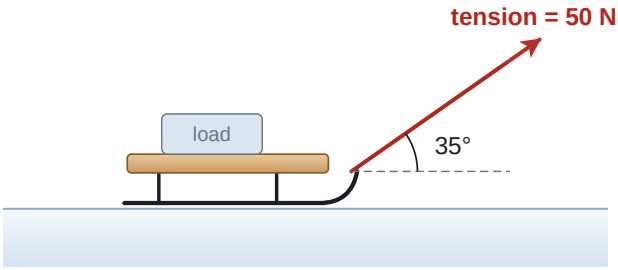


Fig. 18.1

Calculate

(i) the horizontal component of the tension,

horizontal component = ... N **[1]**

(ii) the vertical component of the tension.

vertical component = ... N **[1]**

(c) A drone moves with a horizontal velocity component of 15 m s^{-1} and a vertical (upward) velocity component of 6.0 m s^{-1}, as shown in Fig. 18.2.

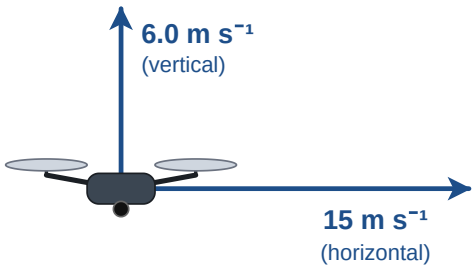


Fig. 18.2

Combine the two components to determine the resultant velocity of the drone. Give its magnitude and the angle it makes with the horizontal.

magnitude = ... m s^{-1}

angle to horizontal = ... ° **[3]**

[Total: 6]

19 **(a)** State Coulomb's law. **[1]**

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(b) Three point charges A, B and C are fixed in a straight line, as shown in Fig. 19.1.

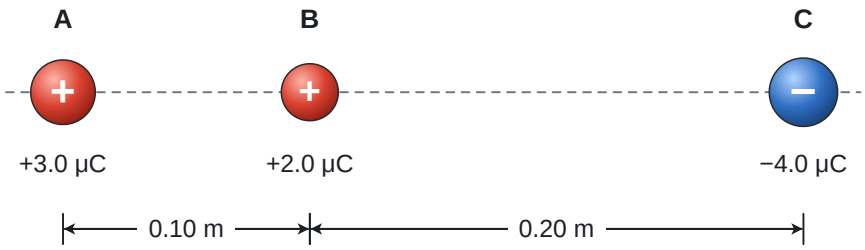


Fig. 19.1

(i) Calculate the magnitude of the force on B due to A.

force due to A = ... N **[1]**

(ii) Calculate the magnitude of the force on B due to C.

force due to C = ... N **[1]**

(iii) Determine the magnitude and the direction of the resultant force on B.

resultant force = ... N

direction: ..... **[1]**

Question 19 continues on the next page.

(c) Three point charges are fixed at the corners X, O and Y of a right-angled triangle, as shown in Fig. 19.2. The angle at O is 90°.

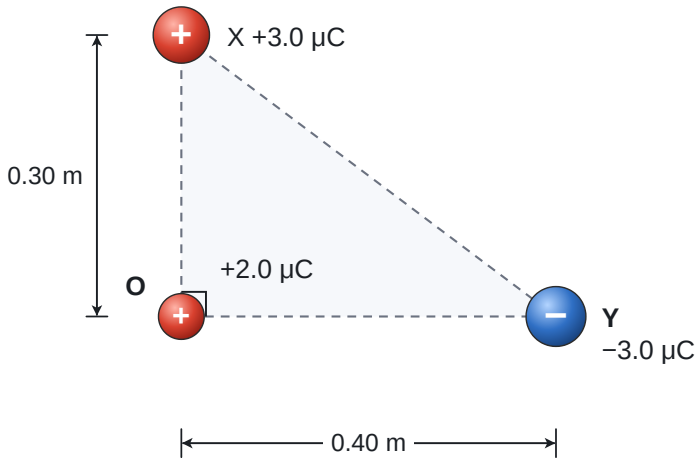


Fig. 19.2

(i) Calculate the force on the charge at O due to the charge at X, and the force on the charge at O due to the charge at Y.

force due to X = ... N

force due to Y = ... N **[1]**

(ii) On Fig. 19.2, draw arrows to show the directions of these two forces on the charge at O. Then determine the magnitude of the resultant force on the charge at O and the angle it makes with the line OY.

resultant force = ... N

angle with OY = ... ° **[2]**

[Total: 7]

Section C · Practical skills

1 question · 10 marks

20 A student investigates the motion of a ball rolling down a straight ramp. The ball is released from rest at the start line and the student uses a stopwatch to measure the time t for the ball to roll a distance d to the finish line. The finish line is moved and the experiment repeated for different values of d. Each time is measured three times.

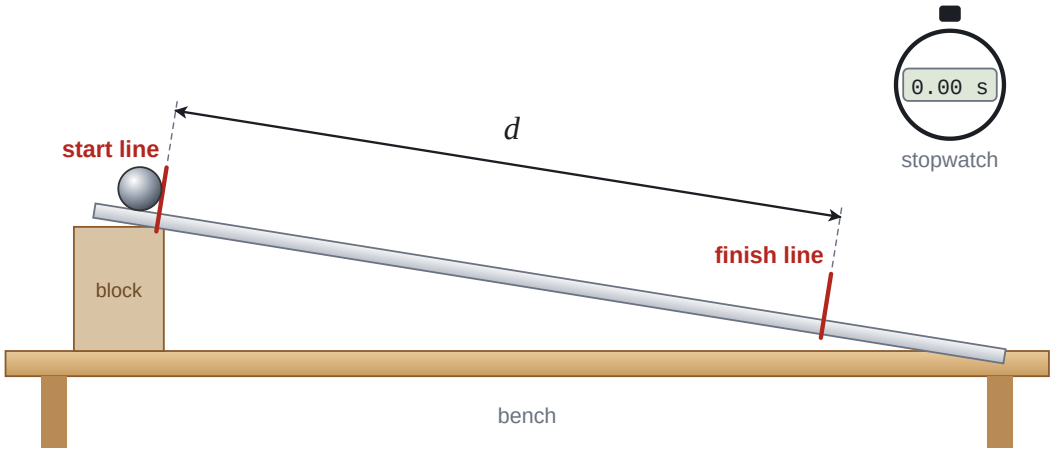


Fig. 20.1

The ball moves with constant acceleration a, so d and t are related by $d = \frac{1}{2}at^2$. The student's results are shown in Table 20.1.

d / m	t_1 / s	t_2 / s	t_3 / s	mean t / s	t^2 / s^2
0.20	0.80	0.84	0.82	0.82	0.67
0.40	1.17	1.13	1.15	1.15	1.32
0.60	1.40	1.44	1.39	1.41	1.99
0.80	1.62	1.66	1.61		
1.00	1.84	1.80	1.85		

Table 20.1

(a) Complete Table 20.1 by calculating the mean value of t and the value of t^2 for d = 0.80 m and d = 1.00 m. **[2]**

(b) On the grid in Fig. 20.2, plot a graph of d on the y-axis against t^2 on the x-axis. Draw the straight line of best fit. **[2]**

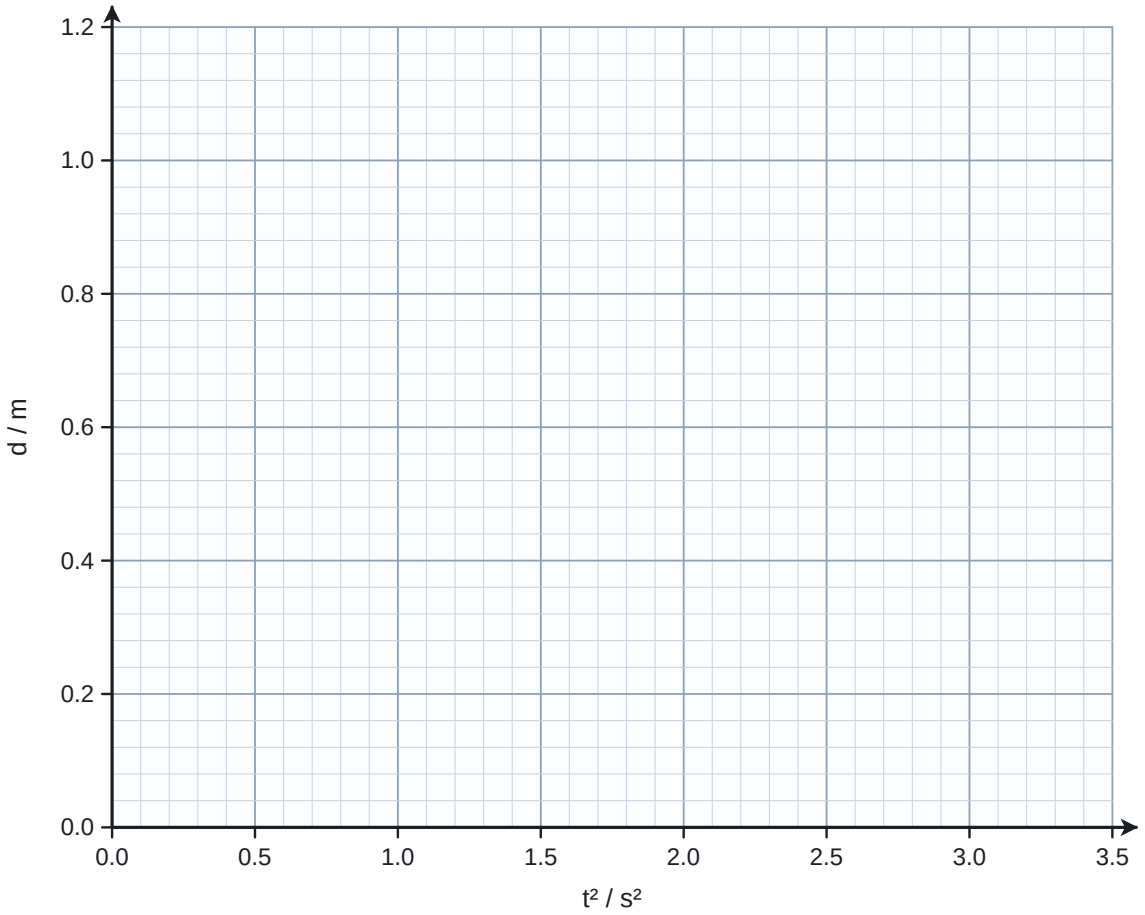


Fig. 20.2

(c) Determine the gradient of your line. Show clearly the values you use.

gradient = ... m s^{-2} **[2]**

(d) Use your answer to (c) and the equation $d = \frac{1}{2}at^2$ to determine the acceleration a of the ball.

a = ... m s^{-2} **[1]**

(e) For d = 0.80 m, the student takes the uncertainty in t to be half the range of the three readings.

(i) Determine the absolute uncertainty in t for d = 0.80 m.

uncertainty = ... s **[1]**

(ii) Calculate the percentage uncertainty in the mean value of t for d = 0.80 m.

percentage uncertainty = ... % **[1]**

(f) Measuring t with a hand-held stopwatch is a source of uncertainty. Suggest **one** improvement to the method that would make the measurements of t more reliable. **[1]**

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[Total: 10]

END OF EXAMINATION